

# Knowledge transfer in academia: an exploratory study on the Not-Invented-Here Syndrome

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**Abstract** Today's society is often claimed to be the “knowledge society”. Knowledge transfer plays a pivotal role in the whole economic system, influencing innovation management in its very core. This study deals with one of the barriers that might hamper knowledge transfer: The Not-Invented-Here Syndrome (NIH). Until now, this phenomenon was mainly analyzed in an industrial context. However, as the universities play an increasingly important role in the knowledge society, the time seems ripe to analyze this barrier of knowledge transfer in an academic context. This paper examines the influence of the research discipline, the scientific output, the attitude towards basic science and the project experience on NIH by analyzing a sample of 166 Austrian professors from the fields of physics and engineering. While we cannot find a significant impact of the first two aspects, we can demonstrate that the latter two do have a significant influence on NIH. Summing up, this study supports results from the existing literature on NIH and sheds light on new aspects providing deeper insights into the comprehension of this complex phenomenon.

**Keywords** Not-Invented-Here Syndrome · University · Collaboration · Innovation · Knowledge transfer · Physics · Engineering

**JEL Classification** D01 · D83 · L32 · O31 · O32 · O33

## 1 Introduction

Innovations are a main driver of economic success and growth (Schmookler 1966; Hornung 2002). However, companies frequently face the question where the much needed innovations should come from. Especially in the early phases of the NPD process, a reasonable alternative to industrial companies are public research institutions like universities (Mansfield and Lee 1996). Mainly focusing on basic and applied research, these

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institutions often serve as the source of ground-breaking ideas and inventions, affecting industry, consumers, and the society as a whole. Examples can be found in nearly every research field: GATORADE<sup>®</sup> (University of Florida) revolutionized the world of sports, whereas magnetic resonance tomography (University of New York/University of Nottingham) provided new opportunities for medical diagnosis and treatment. Especially in the natural and engineering sciences, academic research is conducted at the frontier of scientific progress and can be a valuable source of knowledge.

In his seminal work, Nonaka (1994) explains that knowledge itself is generated by single individuals. To create innovations, this new knowledge has to be transferred to the organizational level. Nonaka illustrates this as an upward spiral in which knowledge undergoes four different modes of conversion: socialization, externalization, internalization and combination. Starting with the individual generating tacit knowledge and subsequently passing through these four modes, knowledge becomes useful for the whole organization at the end of the process, sometimes also reaching the inter-organizational level. Taking a look at this knowledge development, it becomes obvious that knowledge transfer is an indispensable and frequent step in this procedure, especially when keeping in mind the increasing amount of interactions between individuals and groups at the upward end of the spiral.

When knowledge transfer is disturbed or hampered, the complete knowledge generation process may fail. Knowledge transfer barriers are manifold and likewise, several different classifications exist in literature (e.g. Szulanski 1996; Husted and Michailova 2002; Greiner and Franza 2003; Rosen et al. 2007). Generally, three main categories can be identified in most of the studies: barriers concerning environmental aspects, ranging from characteristics of the knowledge itself to a high heterogeneity of team members, barriers on the knowledge provider's side and barriers on the knowledge receiver's side. Especially the latter category has often been neglected in research so far. Being as important as the other two perspectives for the success of knowledge transfer, we focus on that side of the coin. The most important obstacle in the recipient category is the so called Not-Invented-Here Syndrome (NIH).

The NIH Syndrome is defined as "a negative attitude to knowledge that originates from a source outside the own institution" (Lichtenthaler and Ernst 2006, p. 368). This attitude can be observed on the level of the individual. Accordingly, Mehrwald (1999) integrated NIH in a psychological context, describing it as a prejudice against external technology. The existence of this prejudice can lead to severe consequences. External technologies may be evaluated wrongly (Mehrwald 1999) or its implementation may fail afterwards as people attack these technologies in an inadequate manner (Clagett 1967). Moreover, the performance of project teams may decline (Katz and Allen 1982) or the projects themselves can be delayed (de Pay 1989). All in all, NIH affects the complete knowledge cycle in or between organizations, leading to severe disturbances in knowledge transfer. In general, the concept of NIH is widely accepted in research and practice (Mehrwald 1999), although empirical studies examining this phenomenon are still rare (Clagett 1967; Katz and Allen 1982; de Pay 1989; Boyens 1998; Mehrwald 1999; Herzog 2008). However, all of these existing studies focus on industrial researchers as the subjects of analysis. Just in recent years, the importance of academic R&D for innovation became also apparent in literature. Thus, academic NIH is still a white spot on the research map. Therefore, this paper aims to fill this gap and examines the existence of NIH in academia. We further explain the influence of research direction and research discipline. Additionally, we analyze the impact of project experience and scientific output on the extent of NIH.

The remainder of this paper is structured as follows. In the next section, the antecedents and consequences of NIH are described. Taking into account special characteristics of academic R&D, the phenomenon is transferred to academia. Subsequently, hypotheses are deduced and NIH is operationalized. Based on the existing studies, constructs are created and enriched by own considerations. Finally, the presentation of the research results follows before the paper closes with a critical discussion and a conclusion.

## 2 The NIH Syndrome

The NIH Syndrome itself is not inherent to a person by birth but develops over time (Clagett 1967). Several antecedents of the phenomenon are explained in the abovementioned studies. Analyzing these studies more closely, three major antecedents emerge: First of all, human beings generally strive for security (Thomas 1991). Any changes (e.g. external knowledge integration) increase the level of uncertainty and thus make most people feel uncomfortable. By opposing or rejecting external ideas, a status quo can be maintained. Therefore, the will to reduce insecurity can be regarded as one of the major antecedents of NIH (Clagett 1967; Katz and Allen 1982; Boyens 1998; Mehrwald 1999). Secondly, security can be retained by introducing work routines. Often, these routines are disturbed by the confrontation with external knowledge. Consequently, people habitually resist these changes (Clagett 1967; Katz and Allen 1982; Boyens 1998; Mehrwald 1999). Finally, the integration of external knowledge may injure a research group's pride. People may feel affronted when external ideas are regarded as superior to their own (Clagett 1967; Katz and Allen 1982; Boyens 1998; Mehrwald 1999; Herzog 2008).

By definition, NIH infection leads to an incorrect evaluation of external knowledge and a consequential suboptimal use of external ideas (Mehrwald 1999). This misjudgment further often results in a poor project performance and a failure of knowledge integration (Clagett 1967; Katz and Allen 1982). As a third aspect, de Pay (1989) deduces an extension of innovation time and thus an increase in costs. In conclusion, it can be stated that the NIH Syndrome worsens the performance of knowledge integration and impedes knowledge transfer in general.

Antecedents and consequences leading to or resulting from NIH may be different in various environments. As research conditions in industry and academia differ with reference to various aspects like long-term orientation, goals or reward system, it is necessary to consider specific characteristics of university research (López-Martínez et al. 1994; Cyert and Goodman 1997; Amabile et al. 2001). Concerning the antecedents of NIH, injury of group pride takes a unique position. One of the main incentives for academic scientists is the acknowledgement by peers (Siegel et al. 2007; Macho-Stadler et al. 2008). Therefore, the public reputation of the research group or of the university itself is essential. Another aspect heads into the same direction: the fear of knowledge spillovers. In academic research, generally only the first who publishes new results receives credit (Stephan 1996; Stern 2004). Knowledge spillovers may give others a head start in the race for scientific success and reputation.

The consequences of NIH in academia are predominantly comparable to those in industrial research. As NIH leads to a biased evaluation of external knowledge, NIH-infected scientists slow down knowledge dissemination. This can either be caused by questioning external research results in an inappropriate manner or by concentrating on internal resources only. Both might lead to a prolongation of the research process itself. In the context of project work, project failure may be the result. Additionally, mistrusting

external knowledge can lead to a lower level of motivation to integrate it into own work routines. Another important consequence of NIH in academia is the biased self-assessment of the own research group. As group pride is offended by external ideas, one might overestimate the own performance or derogate the external ideas to protect self-esteem (Kahn and Ryen 1972; Branscombe and Wann 1994).

One general difference between the two worlds of research is the monetary aspect, which plays a more pivotal role in industry. This matter affects primarily the consequences of NIH. In an industrial context, project failures may lead to direct monetary losses, whereas in academia such failures of knowledge integration often do not result in direct economic disadvantages. Nevertheless, long-term effects may be serious. A lower knowledge integration performance might lead to lower scientific output. Number of publications can be regarded as an indicator for industry to find academic partners (Elfenbein 2007; Gonzalez-Brambila and Veloso 2007). Thus, lower scientific output may result in indirect monetary effects as for example less third-party funds.

### 3 Hypotheses

#### 3.1 Research field and attitude towards science

Academic research fields cover a wide range of knowledge creation (Un et al. 2010). It is one of the strengths of universities that the spectrum of knowledge is usually wider and deeper than that of industrial or other research facilities (Un et al. 2010). It covers both fundamental and applied research, a distinction that is generally drawn by scientists (Mansfield 1980; Henard and McFadyen 2005). Especially in the 19th century, the former was regarded as “a dedicated scientific pursuit of natural phenomena without seeking any practical application” (Chin-Dusting et al. 2005, p. 891). This kind of pure or basic research is still widely spread today (Chin-Dusting et al. 2005). In contrast, Sir Francis Bacon (1620) claimed in his seminal work that “the real and legitimate goal of the sciences is the endowment of human life with new inventions and riches”. Regardless of whether one individually favors the one or the other, there are some research disciplines that are more application-oriented than others. We suppose that having a close relationship to industry or showing vivid patent and licensing activities are two exemplary indicators for applied science orientation. Biotechnology is one example of such an application-oriented research field, cooperating heavily with industry (McMillan et al. 2000; Poyago-Theotoky et al. 2002; Renault 2006). It is similar to the engineering sciences, which also tend to have tight connections to practice (Thursby and Kemp 2002). In contrast, physics is an example for a research discipline that is more oriented to basic science (Thomas 1987). If one takes a look at the familiar continuum of academic research fields with basic science on the left end and applied on the right end (Stokes 1997), biotechnology and engineering will take a position somewhere on the right. In contrast to this, research fields like physics or mathematics as typical basic sciences will be located more to the left end.

Based on this background and on the distinction between basic and applied science, one might argue that the researchers in each field also differ in their behavior. Landry et al. (2007) also found in their study that engineers show higher knowledge transfer activities than physics. But, we agree with the existing literature that the Not-Invented-Here Syndrome is a psychological construct, representing an attitude towards external knowledge (Mehrwald 1999; Herzog 2008). Attitudes are created in the mind of an individual (Olson and Zanna 1993). Although there are numerous theories about the formation of attitudes,

personal issues like conditioning or heritability are wide-spread as explanatory approaches (Olson and Zanna 1993). However, research discipline is a too superficial criterion as it does not take personal aspects like experiences into account. Thus, being a highly complex psychological concept, we contend that the difference in research discipline should not influence the existence of NIH significantly. Therefore, we hypothesize:

**Hypothesis 1** Academic researchers in the fields of physics and engineering do not differ with regard to the NIH Syndrome.

Assuming that hypothesis 1 holds true and environmental factors do not show a strong impact on NIH, the personal attitude of a researcher may be another explanation for his thoughts on externally sourced knowledge. While universities were traditionally responsible for research and education, they are increasingly expected by policymakers and public authorities to fulfill the third mission of technology transfer (Etzkowitz et al. 2000; Fabrizio 2006; Rafferty 2008). In this light, discussions emerged if industry begins to dictate research agendas and if academic freedom is at stake (Krimsky and Ennis 1991; Monbiot 2003). Accordingly, especially applied research might drift to a kind of contract research. This industry-sponsored research often leads to secrecy agreements as companies aim to gain exclusive rights on the results (Blumenthal et al. 1996). Beyond that, industry as being more short-term oriented often sets tight time schedules (Betz 1997; Meyer-Krahmer and Schmoch 1998). Facing this time pressure, academic applied researchers often need to speed up research efforts. External knowledge can help to accelerate their work and is, therefore, welcomed by applied scientists.

On the other hand, basic researchers are usually not exposed to these time constraints (Cyert and Goodman 1997). But, simultaneously, academic competition influences their behavior. “It is important to stress that recognition in science depends on being first” (Stephan 1996, p. 1202). Eponym is one of the top goals of researchers (Stephan 1996). The lower the number of names attached to a scientific discovery, the easier to remember. Additionally, collaborative research can harm the visibility of a single scientist as he might disappear in the anonymity of the group (Beaver and Rosen 1978). Hence, doing research on one’s own helps to achieve greater reputation among peers (Beaver and Rosen 1978). Combining these two streams of thoughts, we argue that application-oriented researchers often need external knowledge to meet time constraints. Moreover, basic scientists may isolate themselves to prevent knowledge spillovers. Thus, they might be skeptical concerning external knowledge and knowledge exchange in general. Therefore, we deduce the following hypothesis:

**Hypothesis 2** Academic researchers who prefer the principles of applied science are less likely to be infected with the NIH Syndrome.

### 3.2 Project experience and scientific output

As literature treats NIH as a syndrome or virus (Katz and Allen 1982; Mehrwald 1999; Chesbrough 2003, 2006; Lichtenthaler and Ernst 2006; Newman 2007; Herzog 2008), there must be means to cure this infection. Mehrwald and Herzog argue that negative experiences or a general lack of experience with prior projects, being one form of knowledge sharing situation, facilitate NIH (Mehrwald 1999; Herzog 2008). Human beings generally tend to avoid depressing situations (Foa and Kozak 1986). As people head for security, new circumstances can provoke fears and uneasiness (Thomas 1991). Learning

through frequent exposure to these new situations makes the person more familiar with it and thus helps to relieve fears (Foa and Kozak 1986; Thomas 1991).

Projects often are very different, for example, with respect to the partners themselves, especially in heterogeneous collaborations. Cyert and Goodman (1997), among others, state that cooperation between university and industry may harbor essential risks of failure (Hall et al. 2001). These risks are predominantly grounded in a different culture, particularly meaning different time horizons or goals (Jones 2006). Such obstacles can be neglected in inter- and intra-university partnerships because, in general, academic culture is similar (Amabile et al. 2001). However, there are also analogies between university-industry collaboration and purely academic partnerships. Collaborating means working together with external persons outside the own research group (Kraut et al. 1987). Additionally, these projects regularly have a certain time line which entails coordination activities (Kraut et al. 1987). Furthermore, collaboration projects with external partners entail a different kind of communication than internal work routines (Katz and Tushman 1979, 1981).

Doing many projects makes the unfamiliar situation more familiar to the scientist. It allows him to get a certain routine in handling these circumstances. Summarizing these thoughts, both, university-industry and intra-academic projects, should help to prevent or remedy the NIH Syndrome. Therefore, we hypothesize:

**Hypothesis 3** Academic researchers who have been involved in more projects are less likely to be infected with the NIH Syndrome.

In recent years, the amount of publications in science increased steadily (Persson et al. 2004). This development is accompanied by a rising number of co-authorship papers (Wuchty et al. 2007). Science becomes more specialized and the “burden of knowledge” increases, both forcing scientists into more narrow fields of expertise (Jones 2009). Nowadays, universal scholars like Galileo Galilei and Gottfried Wilhelm Leibniz do hardly exist anymore. But even inside one discipline, research fields divide more and more. Accordingly, to understand and finally publish complex research matters collaboration seems to be necessary (Wuchty et al. 2007). Thus, we assume that scientists with a strong track record in publications have to join many collaborative activities. In general, similar obstacles as already explained in the derivation of hypothesis 3 may occur here. Hence, we argue analogously that the more collaborative work is done (seen in the number of publications), the more researchers get familiar with these situations and the lower their NIH infection is. Therefore, we develop the following hypothesis:

**Hypothesis 4** Academic researchers who show a more frequent publication behavior are less likely to be infected with the NIH Syndrome.

## 4 Research design and operationalization

As the variety of knowledge in universities is enormous and hard to cover in one single study, we take physics and engineering sciences as examples for our analysis. These research disciplines cover both sides of the basic and applied research continuum. Several of the existing studies on NIH choose organizations like companies (Boyens 1998), business units (Herzog 2008) or project teams (Katz and Allen 1982) as the level of analysis. These macro perspectives do not fit the academic context. Universities are too divers to serve as a homogenous level of analysis. The same applies to departments as they

generally just subsume all research fields of a single discipline, but only forming a very loose network. Another possible perspective would be that of the research team. In the German university landscape, the professor is often the only senior researcher in his group whereas the rest of the team consists of PhD students. As most of these young researchers only stay shortly in academia, their answers would bias the results. Therefore, our study took the perspective of the single senior researcher (professor) to achieve well-grounded insights and preserve a homogenous sample. Our study addressed 651 Austrian professors from the fields of physics and engineering, affiliated to all universities and polytechnics, in a total population survey. The invitation to a standardized online questionnaire was sent in a personalized email. This questionnaire underwent a pretest by a small group of peer academic researchers. Based on their suggestions, some questions were eliminated or reformulated. The online survey was open for 3 weeks with a reminder sent after 2 weeks. Overall, 601 of the addressees were available; 207 of them answered the survey, representing a response rate of 34.44%. In total, 168 of these questionnaires were filled out completely. Using Mahalanobis distances, three outliers could be detected (Hair 2006). Taking a closer look at these participants, two of them had to be deleted due to pattern-like answers. The final sample for our analysis consists of 166 professors. A possible non-response bias was tested for by splitting the sample into two groups: early respondents (first 75%) and late respondents (last 25%) (Li and Calantone 1998). The rationale underlying this procedure is that late respondents are similar to non-respondents (Armstrong and Overton 1977). Using a Mann–Whitney-*U*-test, no significant differences between the two groups were found. Thus, we can assume that a non-response bias does not play a considerable role.

Mehrwald (1999) was the first to operationalize the NIH Syndrome in a detailed and comprehensive way (Lichtenthaler and Ernst 2006). As his work is embedded in a corporate context, doubts about the applicability of his constructs in an academic framework could be raised. Firstly, items and constructs focus on external technology and not on external knowledge in general. Secondly, the academic self-administration principle is not considered. In academia, principle researchers are generally responsible for both research and management activities. Lastly, Mehrwald analyzes NIH in a market-driven environment, in which product development and market success are key goals. Accordingly, several items are not suitable for academics. To set up a comprehensive item battery, we borrow some items from Mehrwald's work and add own items as well.

Subsequently, we conducted an exploratory factor analysis with promax rotation. To secure a higher stability of the solution, we used a different but analogue sample of 170 German professors for cross-validation (Byrne 2001). This sample was generated in another study using a similar questionnaire and also addressing physicists and engineers. Implementing this step, we can conduct our confirmatory factor analysis with the Austrian sample only. To assess statistical significance, factor loadings should be above .50 with reference to our sample size (Hair 2006). On the other hand, cross-loadings between different items should not exceed .40 (Hair 2006). Items not fulfilling these requirements are checked for their contribution with regards to content. If possible, these items are eliminated to secure a parsimonious and well-grounded solution. Finally, the data can be reflected in a four factor solution:

- Reluctance to knowledge sharing  
Knowledge exchange can also be risky for the own work
- Preference of internal knowledge  
Internally generated knowledge is valued much higher than external knowledge



- Reluctance to external collaborations  
It is absolutely necessary to collaborate with external partners
- Competitive importance of internal knowledge  
The ability to generate knowledge internally is very important to weather the academic competition

The explanations of the different constructs show an example for high scores on the scale. The reliability of the measures in the Austrian sample was evaluated by using Cronbach's Alpha (Hair 2006) (see "Appendix" for overview of factors and quality criteria). We further employed a confirmatory factor analysis to validate our model. Although multivariate normal distribution was not given, we used maximum likelihood as the procedure of choice. West et al. (1995) argue that, as long as skewness and kurtosis do not exceed values of two and seven, respectively, maximum likelihood can be regarded as a robust technique providing reasonable results. The global fit indices for the model were satisfying (comparative fit index (CFI) = .929; root mean square error of approximation (RMSEA) = .069; standardized root mean residual (SRMR) = .054) (Bühner 2006). Concerning the quality criteria in the confirmatory analysis, we based our analysis on the study of Mehrwald and the cut-off criteria used there. These included item reliability, factor reliability, average variance extracted, a significance test of the factor loadings and the Fornell–Larcker criterion.

Reflecting the consequences of the NIH Syndrome, we suppose that it has a significant impact on the abovementioned factors and their parameter values. Therefore, we conducted a confirmatory factor analysis and developed another reflective model with NIH as a higher-order construct. This model again fulfils all relevant quality criteria (CFI = .914; RMSEA = .073; SRMR = .062), being equivalent to those of the first-order factors (Byrne 2001; Hair 2006). In this higher-order model, a Heywood case, i.e. one factor showed a negative error variance, occurred. This was remedied by constraining the error variance to zero (Chen et al. 2001). Furthermore, to secure the independence of the results on inflated correlations caused by data aggregation, we tested the four-first-order factors separately. Again, relevant quality criteria were met. Therefore, we can establish a model with the one higher-order construct "NIH" underlying four-first-order factors mentioned above. Thus, we can state that we can measure the degree of NIH infection in academia with the developed model adequately. To assess a certain score for NIH, we used summed scales whenever combining individual items to a single composite measure (Hair 2006). Hence, NIH value can be quantified on a 7-point Likert scale with high scores showing a high infection.

## 5 Analysis and results

In the context of hypothesis 1, we assumed that physics is a more basic science whereas engineering is more applied. To underline this premise, we analyze several descriptive items, which indicate a high or low application orientation. A closer relationship to industry can be regarded as key characteristic of applied science. Therefore, we use filed patents per year, industry experience of the professor and his attitude towards basic science as indicators for research direction. We conducted three Mann–Whitney-*U*-Tests to analyze whether a significant difference between physics and engineering exists. As it can be seen in Table 1, both research disciplines differ significantly in all three aspects. Hence, we can support the abovementioned premise.



**Table 1** Results of the Mann–Whitney-*U*-test

| Indicators/construct           | Scales          | Groups differing research discipline     |                                              |                                 |
|--------------------------------|-----------------|------------------------------------------|----------------------------------------------|---------------------------------|
|                                |                 | Physics<br>( <i>N</i> = 87)<br>Mean (SD) | Engineering<br>( <i>N</i> = 79)<br>Mean (SD) | Mann–Whitney-<br><i>U</i> -test |
| Patents per year               | Open            | 0.15 (0.39)                              | 0.36 (0.60)                                  | <b>.012</b>                     |
| Industry experience            | 1 = No; 2 = Yes | 1.20 (0.40)                              | 1.57 (0.25)                                  | <b>.000</b>                     |
| Attitude towards basic science | 1–7             | 2.89 (1.47)                              | 4.32 (1.68)                                  | <b>.000</b>                     |
| NIH                            | 1–7             | 3.33 (0.85)                              | 3.25 (0.81)                                  | .546                            |

*SD* standard deviation. Bold values illustrate statistical significance

To test hypothesis 1 and similar to our prior approach, we used a Mann–Whitney-*U*-Test to compare the NIH values between both research disciplines (see also Table 1). As supposed in hypothesis 1 there is no significant difference and thus, our hypothesis finds support in the data.

For analyzing the remaining hypotheses, we employed a linear multiple regression model using NIH as the dependent variable. Attitude towards basic science is operationalized as a dichotomic 7-point scale, asking the participants which of the statement at each end of the scale they favor more (for further details see “Appendix”). As control variables we implemented the age of the scientist, his actual type of research (basic = 0 vs. applied = 1), his former industry employment (no = 1 vs. yes = 2) and his research discipline (engineering = 1 vs. physics = 2). The following quality criteria and requirements of the analysis were fulfilled (Hair 2006). Residuals follow a normal distribution and do not show auto-correlation. Additionally, the model prerequisite of homoscedasticity is satisfied and multicollinearity cannot be identified. Therefore, the regression model can be regarded as robust. Table 2 provides descriptive statistics and correlations of the employed variables/constructs in the regression analysis.

The results of the analysis are presented in Table 3. As shown in this table, the first model focused only on the influence of the control variables. In each of the following models, we subsequently inserted one of the independent variables. We could not identify any significant impact of the controls. By including “Attitude towards science” as an

**Table 2** Means, standard deviations (SD) and correlations of employed variables/constructs

| Variables                   | Mean | SD    | 1     | 2       | 3       | 4       | 5       | 6       | 7     |
|-----------------------------|------|-------|-------|---------|---------|---------|---------|---------|-------|
| 1. Age                      | 4.02 | .904  |       |         |         |         |         |         |       |
| 2. Industry experience      | 1.37 | .485  | -.009 |         |         |         |         |         |       |
| 3. Way of research          | .50  | .502  | -.128 | .523**  |         |         |         |         |       |
| 4. Research discipline      | 1.52 | .501  | .146* | -.386** | -.663** |         |         |         |       |
| 5. Attitude towards science | 3.57 | 1.721 | -.039 | .361**  | .521**  | -.370** |         |         |       |
| 6. Number of projects       | 3.22 | 1.331 | .049  | -.012   | .044    | -.059   | .089    |         |       |
| 7. Publications per year    | 6.60 | 5.138 | -.071 | .004    | -.092   | .067    | -.046   | .239**  |       |
| 8. NIH                      | 3.30 | .830  | .016  | -.137*  | -.128*  | .040    | -.229** | -.162** | -.086 |

*N* = 166; correlations are based on Kendall’s Tau; \* significant at  $p < .05$ ; \*\* significant at  $p < .01$

**Table 3** Results of the multiple regression analysis (dependent variable: NIH)

| Steps and variables entered | Model |          |         |         |
|-----------------------------|-------|----------|---------|---------|
|                             | 1     | 2        | 3       | 4       |
| Age                         | .047  | .058     | .071    | .073    |
| Industry experience         | −.156 | −.108    | −.124   | −.126   |
| Way of research             | −.151 | .012     | .008    | .010    |
| Research discipline         | −.121 | −.137    | −.149   | −.149   |
| Attitude towards science    |       | −.340*** | −.314** | −.314** |
| Number of projects          |       |          | −.194** | −.200*  |
| Publications per year       |       |          |         | .020    |
| $R^2$                       | .050  | .125     | .162    | .162    |
| Adj. $R^2$                  | .027  | .097     | .130    | .125    |
| $\Delta R^2$                |       | .074***  | .037**  | .000    |

$N = 166$ ; \*  $p < 0.05$ ; \*\*  $p < 0.01$ ; \*\*\*  $p < 0.001$

influencing factor, the adjusted  $R^2$  increased to .097. The negative sign of beta shows that the more the participants favor the principles of basic science (being on the left end of the dichotomic scale) the higher their NIH value is. In model 3, adding “Number of projects” leads to a reduction of the significant influence of “Attitude towards basic science”. Nevertheless, the adjusted  $R^2$  rose to .130. Again, there is a negative influence of the independent variable on NIH, indicating that a high number of projects leads to a lower NIH infection. In the last model, “Publications per year” negatively contributed to the overall model fit indicator as adjusted  $R^2$  fell to .125. Hence, we can conclude that we found evidence for hypotheses 2 and 3. In contrast to this, hypothesis 4 found no support in the data and has to be rejected.

## 6 Discussion and conclusion

In this paper, we have presented the results of an empirical examination of the NIH syndrome of 166 Austrian university professors. Aiming at a comprehensive analysis, we firstly transferred this phenomenon to an academic framework theoretically. In a second step, we operationalized the phenomenon to test it among 166 Austrian professors from the physics and engineering sciences. NIH is regarded as a psychological construct (Mehrwald 1999), being very complex in nature. This may be a reason why empirical operationalizations are so rare. As the operationalization is, thus, in a very exploratory stage, we could not base our item selection on prior empirical work. Nevertheless, we have been able to identify four distinct factors representing NIH in a reflective second-order model fulfilling all relevant statistical requirements. Concerning the content dimension of the latent variables, we tried to consider various facets of NIH. This starts with the preference for internally generated knowledge and the perception of the professors on how important outsiders regard internal knowledge generation. These two factors are very close at the core of the NIH Syndrome having prejudices against external knowledge. Further, we included two dimensions not being so intuitively connected to NIH: reluctance to collaboration and reluctance to knowledge sharing. The former does not directly deal with knowledge itself. However, contending that external partners are not needed suggests a comparable attitude

towards the external partners' knowledge. The last latent variable takes a similar perspective dealing with problems occurring during external collaboration. We argue that judging these items high may point to a biased view on external knowledge and all its sidelines. However, this first operationalization of NIH in academia is a sound approach and can prepare the ground for future research in this field.

Further, our study takes a closer look at the context of NIH. It supports the findings of Mehrwald (1999) and others that NIH can be regarded as a psychological issue being individual-based. Focussing on two research disciplines, physics and engineering, we showed that both differ in their orientation: physics is a typical basic science, whereas engineering is more application-oriented. Keeping this difference in mind, the indifference in NIH values of the single participants supports the view that the infection with NIH depends on the person itself and not on the research discipline a person works in.

In a second step, this paper provides insights into the antecedents of NIH. Digging deeper into the abovementioned context, we evaluated the impact of the researchers' mindset on science itself. Neglecting the discussions about basic and applied science, we asked the professors which direction science should take in general. There is a significant relationship demonstrating that those researchers regarding science as a technology-generating activity show a lower NIH value. On the one hand, application-oriented scientists often work more closely with industry partners. Hence, time pressure forces them to consider external knowledge as it speeds up the research process. On the other hand, basic scientists work under the umbrella of the academic incentive system in its purest sense. Consequently, recognition in the scientific community is one of the top goals (Stephan 1996). Therefore, they may isolate themselves to avoid knowledge spillovers, leading to a natural reservation against external ideas. Another explanation might be the picture of the lonesome but brilliant researcher. Once, F. Scott Fitzgerald stated in his notes: "No grand idea was ever born in a conference" (Fitzgerald and Wilson 1945, p. 122). Wuchty et al. (2007) argue that individual thinkers often outperform groups in highly complex tasks. Considering basic science as providing groundbreaking ideas at the edge of the knowledge frontier and as the pacemaker for technological progress (Stokes 1997), basic scientists may tend to hermit behavior more often. Another thought supports the hypothesis' result. Researchers favoring the principles of basic research may show a higher group pride. This may evolve as these scientists usually start their academic career directly after their studies. 80.77% of those that favor basic science have never worked in industry, whereas 68.18% of those that regard science as a technology-generating activity have been employed in industry prior to their university career. Hence, a longer tenure regularly leads to higher loyalty and pride (O'Reilly and Chatman 1986; Balfour and Barton 1996). A longer employment leads to a lower turnover intent, implying that the individual accepts and favors the organization and its goals and mission (Balfour and Barton 1996). O'Reilly and Chatman (1986) found in their study that people with longer service show a higher commitment to the affiliation. However, a more detailed analysis showed that this commitment is not based on compliance but on pride (O'Reilly and Chatman 1986). Nevertheless, these considerations call for future research.

Mehrwald (1999) and Herzog (2008) already explained for industrial researchers that project experience diminishes the likelihood of NIH, a finding we can confirm in an academic context. Unfortunately, our study just analyzes the pure number of projects. Future research should also consider the success of the project work. However, we would contend that the success will also influence the NIH value. Researchers participating in a lot of projects should have made fairly good experiences. Otherwise, they would have diminished their number of projects as most individuals try to avoid depressing situations

(Foa and Kozak 1986). Hence, project success may moderate the strength of the effect number of projects on NIH. Nevertheless, to gain more insights into this influencing factor, future research should take a look at the success of the projects.

Lastly, this paper analyzed the impact of publication activity on NIH infection. Surprisingly, we found no support for hypothesis 4. Assuming that a high number of publications indicates a high collaboration activity, a closer look at the authorship of publications may shed more light on this aspect. Future research should consider the kind of co-authorship being either based on an internal collaboration or on an external one. We argue that there is a difference and that researchers heavily publishing with external partners show a lower NIH value.

Concerning limitations, we have to keep in mind that a generalization of results is difficult. On the one hand, the culture of the participants may play an important role. The national culture as well as the academic culture may be different in other countries like e.g. the USA. On the other hand, we just asked two different research disciplines. Although we deliberately chose one typical basic and one typical applied research discipline, some more fields have to be considered. Nevertheless, our first results show that there should be no relationship between discipline and NIH which would allow generalization in that context. To stabilize these results further research should be done on professors from other fields.

Finally, we can derive some practical implications. Firstly, we want to sensitize academic researchers for this phenomenon. Van der Steen and Enders (2008) underline the importance of learning in academia as the environment and the role of universities have changed in recent years. Open discussions about external ideas are fruitful and foster high-quality research. Secondly, it has to be discussed if a strong scientific competition is helpful for academic research. The return to the pristine root of science—knowledge generation—may improve research in its purest sense. Free knowledge dissemination must outplay reservations about results of other members of the scientific community only due to reputation ambition. Admittedly, a critical and objective assessment of all research results may not be neglected in this context. Thirdly, managers planning collaboration with universities should keep potential barriers of knowledge transfer in mind. The cultural gap between industrial researchers and their academic counterparts may be even deeper, when academics do not show a long project record and are more basic researchers. Both foster NIH development and therefore may hamper the willingness to consider external opinions in an adequate manner and thus may impede knowledge transfer.

## 7 Conclusion

Knowledge and technological progress are the main drivers of economic growth in modern societies (Duderstadt 2006; Weber 2006). Especially in Western countries, which are often poor in raw materials, the stock of knowledge is critical for their competitive advantage. In these 'knowledge societies', the role of universities is redefined, pushing academia in a central position (Mansfield 1991). As knowledge transfer is essential for the good of society this paper investigates one of the barriers that may hamper it in academia: the Not-Invented-Here Syndrome.

To the best of our knowledge, this paper is the first to operationalize NIH in an academic context. Developing a second-order construct with four-first-order factors, the model represents various facets of NIH. This provides a detailed and comprehensive overview of this phenomenon. Additionally, we were able to identify some antecedents of NIH in academia. The attitude towards science and the number of projects showed a

significant influence on NIH. The more professors regard science as a technology-generating activity, the lower is their NIH value. Analogously, the more research projects faculty have done, the lower their NIH value is. In contrast, the number of publications per year does not show any impact on NIH. Moreover, we confirmed the view of Mehrwald that NIH is a psychological construct on an individual basis being independent of the research discipline.

All in all, we believe our study extends the theory of NIH and provides new insights into academic knowledge transfer. However, we have to admit that our regression model explains a fairly small amount of variance of the dependent variable. But as NIH is a psychological multi-layered construct it would be presumptuous to think that a complete explanation is possible. Future research should dig deeper into the components of NIH and investigate more possible influencing factors. Nevertheless, we believe that our study enhances the understanding of NIH, especially in an academic context and contributes to the existing body of literature on knowledge transfer.

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## Appendix

See Tables 4, 5 and 6.

**Table 4** Constructs, items, factor loadings, Cronbach's Alphas

| Questionnaire items                                                                                                                                              | Factor loadings |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <i>Reluctance to knowledge sharing</i> (3 items, Cronbach's Alpha = 0.571)<br>[1 = strongly disagree; 7 = strongly agree]                                        |                 |
| 1. The risk that own technological knowledge can be disclosed during collaboration with the knowledge provider makes external knowledge sourcing uncomley for us | .845            |
| 2. External collaboration disturbs the internal working situation                                                                                                | .750            |
| 3. Due to the exchange of research results with other research teams I fear that these teams build up core competencies in this field                            | .608            |
| <i>Preference for internal knowledge</i> (3 items, Cronbach's Alpha = 0.696)<br>[1 = strongly disagree; 7 = strongly agree]                                      |                 |
| 1. We rather develop knowledge on our own than being dependent on the knowledge provider's cooperation in order to understand the external knowledge             | .850            |
| 2. The risk of failure of an internal research project is lower than the risk that the integration of externally sourced knowledge into internal processes fails | .781            |
| 3. One should always have less confidence in externally developed knowledge than in internally developed knowledge                                               | .738            |
| <i>Reluctance to external collaborations</i> (2 items, Cronbach's Alpha = 0.574)<br>[1 = strongly disagree; 7 = strongly agree]                                  |                 |
| 1. External collaborations are absolutely necessary to survive in academic competition. (R)                                                                      | .838            |
| 2. In many research projects we are reliant on the collaboration with external partners. (R)                                                                     | .838            |

**Table 4** continued

| Questionnaire items                                                                                                                    | Factor loadings |
|----------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <i>Competitive importance of internal knowledge</i> (2 items, Cronbach's Alpha = 0.585)<br>[1 = strongly disagree; 7 = strongly agree] |                 |
| 1. Our reputation in the scientific community is significantly based on our ability to develop knowledge internally                    | .842            |
| 2. Our reputation in practice is significantly based on our ability to develop knowledge internally                                    | .842            |
| <i>NIH</i> (4 constructs, Cronbach's Alpha = 0.604) [1 = strongly disagree; 7 = strongly agree]                                        |                 |
| • Reluctance to knowledge sharing                                                                                                      | .692            |
| • Preference for internal knowledge                                                                                                    | .849            |
| • Reluctance to external collaborations                                                                                                | .503            |
| • Competitive importance of internal knowledge                                                                                         | .674            |

$N = 166$ ; ( $R$ ) = reverse coded items; Exploratory factor analysis was performed using SPSS 16.0; Confirmatory factor analysis was performed using AMOS 7; Item values were standardized to run analysis; Goodness-of-fit for the overall measure model are: CFI = 0.914; RMSEA = 0.073; SRMR = 0.062

**Table 5** Means, standard deviations (SD) and correlations of items from the German sample employed in exploratory factor analysis

| Variables                                         | Mean | SD    | 1      | 2      | 3      | 4      | 5      | 6    | 7      | 8    | 9 |
|---------------------------------------------------|------|-------|--------|--------|--------|--------|--------|------|--------|------|---|
| 1. Reluctance to knowledge sharing 1              | 3.05 | 1.545 |        |        |        |        |        |      |        |      |   |
| 2. Reluctance to knowledge sharing 2              | 2.33 | 1.426 | .288** |        |        |        |        |      |        |      |   |
| 3. Reluctance to knowledge sharing 3              | 2.85 | 1.406 | .409** | .477** |        |        |        |      |        |      |   |
| 4. Preference for internal knowledge 1            | 3.42 | 1.568 | .295** | .202** | .333** |        |        |      |        |      |   |
| 5. Preference for internal knowledge 2            | 3.84 | 1.568 | .228** | .268** | .257** | .469** |        |      |        |      |   |
| 6. Preference for internal knowledge 3            | 4.02 | 1.544 | .094   | .126   | .135   | .329** | .358** |      |        |      |   |
| 7. Reluctance to external collaborations 1        | 2.85 | 1.922 | -.093  | .120   | .055   | -.004  | .190*  | .033 |        |      |   |
| 8. Reluctance to external collaborations 2        | 2.77 | 1.554 | -.010  | .219** | .117*  | .009   | .157*  | .034 | .410** |      |   |
| 9. Competitive importance of internal knowledge 1 | 4.61 | 1.718 | .219** | .065   | .036   | .190*  | .156*  | .096 | .123   | .110 |   |

**Table 5** continued

| Variables                                          | Mean | SD    | 1      | 2    | 3     | 4      | 5      | 6      | 7     | 8     | 9      |
|----------------------------------------------------|------|-------|--------|------|-------|--------|--------|--------|-------|-------|--------|
| 10. Competitive importance of internal knowledge 2 | 4.48 | 1.547 | .275** | .044 | .178* | .275** | .242** | .356** | -.107 | -.063 | .332** |

$N = 170$ ; correlations are based on Pearson; \*significant at  $p < .05$ ; \*\* significant at  $p < .01$

**Table 6** Items

#### Questionnaire items

##### *Attitude towards science*

Which of the two statements about science do you support more? [1 = Science is an idea-generating activity. It is dedicated to explain natural phenomena. Its value lies in theoretical aspects. 7 = Science is a technology-developing activity. It is dedicated to serve mankind. Its value lies in its practical use.]

##### *Industry experience*

Prior to your academic career, have you already worked in industry? [no = 1; yes = 2]

##### *Patents per year*

Please estimate the average number of patents you file per year [free answer]

##### *Publications per year*

Please estimate your average number of publications per year [free answer]

##### *Project experience*

In how many research collaborations did you personally participated in during your career? [1 ≤ 5; 2 = 5–10; 3 = 11–15; 4 = 16–20; 5 ≥ 20]

##### *Age of the researcher*

How old are you? [1 ≤ 30 years; 2 = 30–39 years; 3 = 40–49 years; 4 = 50–59 years; 5 ≥ 59 years]

##### *Research discipline*

What is your research discipline? [1 = engineering; 2 = physics]

##### *Way of research*

Please specify the kind of research you do? [1 = basic science; 2 = applied science]

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